

Quantum Mechanics I Exercise 3

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1 The semiclassical limit for momentum space path integral

1.1 Classical equations and boundary conditions

Following the same method in the notes, we start from the path integrals in phase space formulation:

$$K(x_f, t_f : x_i, t_i) = \int Dx(t) Dp(t) \exp \left[\frac{i}{\hbar} S[p(t), x(t)] \right] \quad (1)$$

With the action

$$S[p(t), x(t)] = \int_{t_i}^{t_f} dt \left[p(t) \frac{dx(t)}{dt} - H(p(t), x(t)) \right] \quad (2)$$

Since we want this in the momentum space:

$$\begin{aligned} K(p_f, t_f : p_i, t_i) &= \langle p_f | U(t_f, t_i) | p_i \rangle = \int dx_i dx_f \langle p_f | x_f \rangle \langle x_f | U(t_f, t_i) | x_i \rangle \langle x_i | p_i \rangle = \\ &= \frac{1}{2\pi\hbar} \int dx_i dx_f e^{-ip_f x_f / \hbar} K(x_f, t_f : x_i, t_i) e^{ip_i x_i / \hbar} = \\ &= \frac{1}{2\pi\hbar} \int dx_i dx_f e^{\frac{i}{\hbar} (p_i x_i - p_f x_f)} K(x_f, t_f : x_i, t_i) \\ &= \frac{1}{2\pi\hbar} \int Dx(t) Dp(t) \exp \left[\frac{i}{\hbar} (S[p(t), x(t)] + p_i x_i - p_f x_f) \right] \end{aligned} \quad (3)$$

The object in the exponent can be simplified by integrating by parts the $p\dot{x}$ in the action:

$$\begin{aligned} \tilde{S}[p(t), x(t)] &= S[p(t), x(t)] + p_i x_i - p_f x_f = \\ &= \int_{t_i}^{t_f} dt \left[p(t) \frac{dx(t)}{dt} - H(p(t), x(t)) \right] + p_i x_i - p_f x_f = \\ &= \int_{t_i}^{t_f} dt \left[-x(t) \frac{dp(t)}{dt} - H(p(t), x(t)) \right] \end{aligned} \quad (4)$$

In the limit $\hbar \rightarrow 0$, the stationary solutions will dominate the integral, which will happen when the exponent is zero:

$$\delta \tilde{S} = 0 \quad (5)$$

We can look at the object inside the time integral as a lagrangian, and so the EOMs that stem from the minimization of the integral are the E-L equations for x and p . In other words, in the phase space x and p are two coordinates of this lagrangian, so we can perform E-L equation for each of them. We will get to the canonical relations:

$$\frac{dx_{cl}(t)}{dt} = \frac{\partial H(p(t), x(t))}{\partial p(t)} \Big|_{p=p_{cl}, x=x_{cl}} \quad \frac{dp_{cl}(t)}{dt} = - \frac{\partial H(p(t), x(t))}{\partial x(t)} \Big|_{p=p_{cl}, x=x_{cl}} \quad (6)$$

Finally, since the path is between predetermined momentum states, they will give the boundary conditions:

$$p_{cl}(t_i) = p_i \quad p_{cl}(t_f) = p_f \quad (7)$$

1.2 The Gaussian Integral

Similarly to the coordinate space derivation, we will define the deviation from the classical momentum:

$$p(t) = p_{cl}(t) + q(t) \quad (8)$$

With the boundary conditions

$$q(t_i) = q(t_f) = 0 \quad (9)$$

And a deviation from the classical path:

$$x(t) = x_{cl} + y(t) \quad (10)$$

Without boundary conditions. We can now rewrite the path integral as

$$\begin{aligned} K(p_f, t_f : p_i, t_i) &= \frac{1}{2\pi\hbar} \int Dx(t) Dp(t) \exp \left[\frac{i}{\hbar} \tilde{S}[p(t), x(t)] \right] = \\ &= \frac{1}{2\pi\hbar} \exp \left[\frac{i}{\hbar} \tilde{S}[p_{cl}(t), x_{cl}(t)] \right] \int Dx(t) Dp(t) \exp \left[\frac{i}{\hbar} \left(\tilde{S}[p(t), x(t)] - \tilde{S}[p_{cl}(t), x_{cl}(t)] \right) \right] = \\ &= F(p_f, t_f : p_i, t_i) \exp \left[\frac{i}{\hbar} \tilde{S}[p_{cl}(t), x_{cl}(t)] \right] \end{aligned} \quad (11)$$

Which means

$$F(p_f, t_f : p_i, t_i) = \frac{1}{2\pi\hbar} \int Dx(t) Dp(t) \exp \left[\frac{i}{\hbar} \left(\tilde{S}[p(t), x(t)] - \tilde{S}[p_{cl}(t), x_{cl}(t)] \right) \right] \quad (12)$$

Expanding $\tilde{S}[p(t), x(t)]$ around $p_{cl}(t)$ and $x_{cl}(t)$, and remembering that still $\delta\tilde{S} = 0$, we get

$$\tilde{S}[p(t), x(t)] = \tilde{S}[p_{cl}(t), x_{cl}(t)] + \frac{1}{2} \delta^2 \tilde{S}[q(t), y(t)] + O(\delta^3 \tilde{S}[q(t), y(t)]) \quad (13)$$

To find the value of the second variation $\delta^2 \tilde{S}[q(t)]$ we need to similarly expand the Hamiltonian. In the following calculation we leave the time dependence implicit:

$$\begin{aligned} H(p, x) &= H(p_{cl}, x_{cl}) + \frac{\partial H}{\partial x} \Big|_{p=p_{cl}, x=x_{cl}} y + \frac{\partial H}{\partial p} \Big|_{p=p_{cl}, x=x_{cl}} q + \\ &+ \frac{1}{2} \frac{\partial^2 H}{\partial x^2} \Big|_{p=p_{cl}, x=x_{cl}} y^2 + \frac{\partial^2 H}{\partial x \partial p} \Big|_{p=p_{cl}, x=x_{cl}} yq + \frac{1}{2} \frac{\partial^2 H}{\partial p^2} \Big|_{p=p_{cl}, x=x_{cl}} q^2 + O(\partial^3 H) = \\ &= H(p_{cl}, x_{cl}) - \dot{p}_{cl} y + \dot{x}_{cl} q + \\ &+ \frac{1}{2} \frac{\partial^2 H}{\partial x^2} \Big|_{p=p_{cl}, x=x_{cl}} y^2 + \frac{\partial^2 H}{\partial x \partial p} \Big|_{p=p_{cl}, x=x_{cl}} yq + \frac{1}{2} \frac{\partial^2 H}{\partial p^2} \Big|_{p=p_{cl}, x=x_{cl}} q^2 + O(\partial^3 H) \end{aligned} \quad (14)$$

We can now open the first part of the exponent using the deviations:

$$\begin{aligned} \int_{t_i}^{t_f} -x \dot{p} dt &= - \int_{t_i}^{t_f} [x_{cl} + y] [\dot{p}_{cl} + \dot{q}] dt = - \int_{t_i}^{t_f} x_{cl} \dot{p}_{cl} + y \dot{p}_{cl} + x_{cl} \dot{q} + y \dot{q} dt = \\ &= - [x_{cl} q]_{t_i}^{t_f} - \int_{t_i}^{t_f} x_{cl} \dot{p}_{cl} + y \dot{p}_{cl} - \dot{x}_{cl} q + y \dot{q} dt = - \int_{t_i}^{t_f} x_{cl} \dot{p}_{cl} + y \dot{p}_{cl} - \dot{x}_{cl} q + y \dot{q} dt \end{aligned} \quad (15)$$

Where in the last transition we used the fact that $q(t_i) = q(t_f) = 0$. These two equation mean that:

$$\begin{aligned} \tilde{S}[p(t), x(t)] &= - \int_{t_i}^{t_f} dt [x_{cl} \dot{p}_{cl} + y \dot{p}_{cl} - \dot{x}_{cl} q + y \dot{q} + H(p_{cl}, x_{cl}) - \dot{p}_{cl} y + \dot{x}_{cl} q + y \dot{q} \\ &\quad + \frac{1}{2} \frac{\partial^2 H}{\partial x^2} \Big|_{p=p_{cl}, x=x_{cl}} y^2 + \frac{\partial^2 H}{\partial x \partial p} \Big|_{p=p_{cl}, x=x_{cl}} y q + \frac{1}{2} \frac{\partial^2 H}{\partial p^2} \Big|_{p=p_{cl}, x=x_{cl}} q^2 + O(\partial^3 H)] = \\ &= \tilde{S}[p_{cl}(t), x_{cl}(t)] - \int_{t_i}^{t_f} dt y \dot{q} + \left[\frac{1}{2} \frac{\partial^2 H}{\partial x^2} \Big|_{p=p_{cl}, x=x_{cl}} y^2 + \frac{\partial^2 H}{\partial x \partial p} \Big|_{p=p_{cl}, x=x_{cl}} y q + \frac{1}{2} \frac{\partial^2 H}{\partial p^2} \Big|_{p=p_{cl}, x=x_{cl}} q^2 \right] + O(\delta^3 \tilde{S}) \end{aligned} \quad (16)$$

Comparing this to equation 13, it's clear to see that

$$\frac{1}{2} \delta^2 \tilde{S}[q(t), y(t)] = - \int_{t_i}^{t_f} y \dot{q} + \frac{1}{2} \frac{\partial^2 H}{\partial x^2} \Big|_{p=p_{cl}, x=x_{cl}} y^2 + \frac{\partial^2 H}{\partial x \partial p} \Big|_{p=p_{cl}, x=x_{cl}} y q + \frac{1}{2} \frac{\partial^2 H}{\partial p^2} \Big|_{p=p_{cl}, x=x_{cl}} q^2 dt \quad (17)$$

1.3 Finding F and the Van-Vleck determinant

Classically, propagating from a given initial momentum p_i at time t_i to a given final momentum p_f at time t_f requires a certain x_i at time t_i . Changing the final momentum requires changing the initial position. Expanding around the classical solution, using the constant probability density for x at a given momentum $\omega(x) dx_i = dx_i / 2\pi\hbar$, the probability to propagate into some momentum interval, $p_f < p < p_f + dp_f$ at t_f will be, by the definition of the propagator,

$$|K(p_f, t_f; p_i, t_i)|^2 dp_f = \frac{1}{2\pi\hbar} dx_i = \frac{1}{2\pi\hbar} \frac{dp_f}{|dp_f/dx_i|} \quad (18)$$

And so, the semiclassical amplitude F is

$$|F(p_f, t_f; p_i, t_i)| = \frac{1}{\sqrt{2\pi\hbar |dp_f/dx_i|}} = \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{dx_i}{dp_f}} \quad (19)$$

Since the exponent follows the canonical relations in equation 6, we can say that

$$x_i = \frac{\partial \tilde{S}}{\partial p_i} \quad (20)$$

This can also be seen from equation 4

$$\frac{\partial \tilde{S}}{\partial p_i} = \frac{\partial}{\partial p_i} \left[\int_{t_i}^{t_f} dt \left[p(t) \frac{dx(t)}{dt} - H(p(t), x(t)) \right] + p_i x_i - p_f x_f \right] = x_i \quad (21)$$

And so

$$|F(p_f, t_f; p_i, t_i)| = \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{\partial^2 \tilde{S}}{\partial p_i \partial p_f}} \quad (22)$$

Generalizing this to multiple dimensions is simple:

$$|K(p(t_f), t_f; p(t_i), t_i)|^2 \prod_{k=1}^M dx_k(t_f) = \prod_{n=1}^M 2\pi\hbar dp_n(t_i) = \det \left[\frac{\partial p_k(t_f)}{\partial x_n(t_i)} \right] \prod_{n=1}^M dx_n(t_i) \quad (23)$$

Finally, using the opposite determinant rule $(\det P)^{-1} = \det(P^{-1})$ and again the canonical relations $x_n(t_i) = \frac{\partial \tilde{S}}{\partial p_n(t_i)}$:

$$|F(p(t_f), t_f, p(t_i), t_i)| = \frac{1}{(2\pi\hbar)^{M/2} \sqrt{\det [\partial p_k(t_f) / \partial x_n(t_i)]}} = \frac{1}{(2\pi\hbar)^{M/2}} \sqrt{\det \left| \frac{\partial^2 \tilde{S}}{\partial p_k(t_f) \partial p_n(t_i)} \right|} \quad (24)$$

1.4 The Green's Function

The Energy dependent Green's function's relation to the propagator in the phase space can be written as

$$G(p_f, p_i; E) = \frac{1}{i\hbar} \int_0^\infty dTK(p_f, T; p_i, 0) e^{iET/\hbar} = \frac{1}{i\hbar} \int_0^\infty dTF(p_f, T; p_i, 0) e^{i(ET + \tilde{S}[p_{cl}, x_{cl}])/\hbar} \quad (25)$$

For a classical trajectory $\hbar \rightarrow 0$, we can again use the stationary phase method, and require the derivative of the exponent by T to be zero:

$$E = -\frac{\partial \tilde{S}[p_{cl}, x_{cl}]}{\partial T} \quad (26)$$

Using equation 4 for a classical path, we get

$$E = -\frac{\partial}{\partial T} \int_0^T dt \left[-x(t) \frac{dp(t)}{dt} - H(p(t), x(t)) \right] = \frac{\partial}{\partial T} \left(\int_{p_i}^{p_f} x dp + E_{cl} T \right) = E_{cl} \quad (27)$$

Which makes the expression in the exponent in the formula for the Green's function

$$W(p_f, p_i; E) = ET + \tilde{S}[p_{cl}, x_{cl}] = ET - \left(\int_{p_i}^{p_f} x dp - E_{cl} T \right) = - \int_{p_i}^{p_f} x dp \quad (28)$$

Which makes the stationary phase approximation of the Green's function

$$G(p_f, p_i; E) \approx F(p_f, T(E); p_i, 0) \left(\frac{2\pi}{i\hbar \gamma(E)} \right)^{1/2} e^{iW(p_f, p_i; E)/\hbar} \quad (29)$$

Where

$$\gamma(E) = \left. \frac{\partial^2 W}{\partial T^2} \right|_{T(E)} = \left. \frac{\partial E_{cl}}{\partial T} \right|_{T(E)} \quad (30)$$

2 Tunneling in momentum space

2.1 Classical trajectories with energy below the barrier

For a one dimensional barrier, the constant energy lines in the phase space look like

$$E = \frac{p^2}{2m} + V(x) \quad (31)$$

where E is constant. This means we can get two plots

$$p = \pm 2m \sqrt{E - V(x)} \quad (32)$$

Where the sign signifies the direction of motion. For a barrier with width A of the structure

$$V(x) = \begin{cases} \varphi(x) & |x| < A/2 \\ 0 & else \end{cases} \quad (33)$$

Where $\varphi(x)$ is some positive function for which $E < \varphi_{max} = V_{max}$. The equal-energy line in the phase space is split into two trajectories, each matching a different side of the barrier, leaving the center, in the area $a < x < b$ such that $V(a) = V(b) = \varphi(a) = \varphi(b) = E$, undefined for real times. For these trajectories, the classical propagation begins at x_i on one side, reaches point a or b , depending on the direction of motion. At that point $p = 0$, and the motion turns back, moving away from the barrier and to x_f which is on the same side, moving with opposite momentum. We can see the equal-energy lines for the barrier $\varphi(x) = -\frac{1}{2}\omega^2 x^2$, in figure 1.

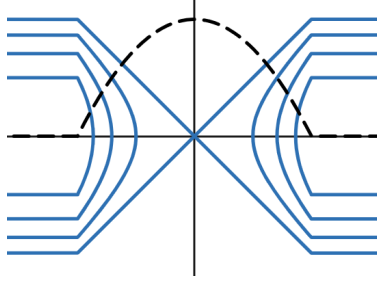


Figure 1: Classical Phase space trajectories for the barrier potential $\varphi(x) = -\frac{1}{2}\omega^2 x^2$ for the energies $E = 0.25\varphi_{max}, 0.5\varphi_{max}, 0.75\varphi_{max}$, and φ_{max} . The plot with $E = \varphi_{max}$ is the one that passes through the center. The dotted line is $V(x)$.

2.2 Imaginary time trajectories with energy below the barrier

Adding the imaginary time trajectory connecting the two parts, it will correspond to moving in imaginary time, which is equivalent to switching the order of the terms in the square root in equation 32:

$$p = \pm 2m\sqrt{V(x) - E} \quad (34)$$

giving the red plots in figure 2. It is clear to see in figure 2 that the instanton trajectory connects to the

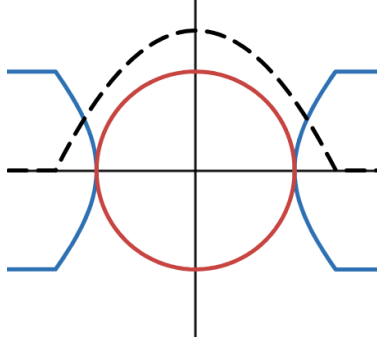


Figure 2: Classical Phase space trajectories for the barrier potential $\varphi(x) = -\frac{1}{2}\omega^2 x^2$ for the energy $E = 0.5\varphi_{max}$ including motion in an imaginary time. The red line is the imaginary time trajectories connecting the two sides of the barrier. The dotted line is $V(x)$.

two classical trajectories at $x = a$ and $x = b$, the points at which $E = V(x)$ on the two sides of the barrier, where the square roots in both equation 32 and 34 are zero. In these trajectories, the particle starts from one side and ends on the other side of the barrier, with the real part of its momentum starting and finishing with the same value.

2.3 Classical trajectories with energy above the barrier

The classical trajectories for particles moving with $E > V_{max}$ can be seen in figure 3. One can see that in each side of the barrier there are two distinct trajectories with opposite momenta that do not interest. In other words, for every 2 points x_i and x_f on the same side of the barrier, there are two different paths with $p = \pm 2m\sqrt{E - V}$ that connect them. This means that the classical trajectories do not contain any option for reflection.

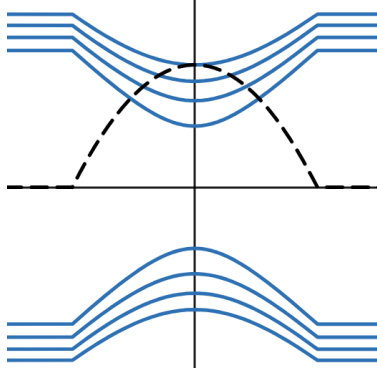


Figure 3: Classical Phase space trajectories for the barrier potential $\varphi(x) = -\frac{1}{2}\omega^2 x^2$ for the energies $1.25\varphi_{max}$, $1.5\varphi_{max}$, $1.75\varphi_{max}$, and $2\varphi_{max}$. The dotted line is $V(x)$.

2.4 Instantons in trajectories with energy above the barrier

Intuitively, we can think about an instanton that will connect that two separate trajectories in the same way the instanton allowed the particle to tunnel through the barrier. Since momentum is never zero, the instanton will have to flip it, and thus the most logical place to put it is along the p axis. The suggested trajectory can be seen in figure 4.

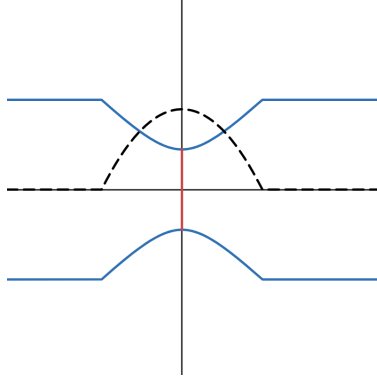


Figure 4: Classical Phase space trajectories for the barrier potential $\varphi(x) = -\frac{1}{2}\omega^2 x^2$ for the energy $1.25\varphi_{max}$ including motion in an imaginary time. The red line is the imaginary time trajectory connecting the two opposite- p trajectories. The dotted line is $V(x)$.

2.5 Green functions

In this and the following section, we will assume, for the sake of simplicity and without loss of generality, that the potential is maximized for $x = 0$. For every trajectory (n):

$$G^{(n)}(x_i, x_f; E) \approx \frac{m}{i\hbar} \frac{1}{\sqrt{p(x_i)p(x_f)}} \exp\left(\frac{i}{\hbar} \int_{x_i}^{x_f} p(x) dx\right) \quad (35)$$

with the integral in the exponent going along the trajectory path. For every trajectory, the path starts by moving from x_i to 0 with momentum p_i and ends by moving back from 0 to x_f which is on the same side as x_i , and with an opposite momentum $p_f = -p_i$.

In the middle of the trajectory $G^{(n)}$ there are $2n + 1$ passages through the complex plane in both p and x . In the complex p plane, in order to change the momentum from p_i to p_f , it will move along the semicircle with a constant $|p|$. In the complex x plane, it will have to pass through a branch cut in the momentum $p = \pm 2m\sqrt{E - V(x)}$ to get the phase in the momentum. So, naming the closest point on the branch cut x_c , the trajectory moves from 0 to the x_c and back:

$$G^{(n)}(x_i, x_f; E) \approx \frac{m}{i\hbar} \frac{1}{\left| \sqrt{p(x_i)p(x_f)} \right|} \exp \left(\frac{i}{\hbar} \left(\int_{x_i}^0 p(x) dx + i(4n + 2) \int_0^{x_c} |p(x)| dx + \int_0^{x_f} p(x) dx \right) \right) \quad (36)$$

2.6 The Tunneling action

The action that we got in the exponent is very similar to the action for tunneling:

$$S[x] = \frac{i}{\hbar} \left(\int_{x_i}^0 p(x) dx + i(4n + 2) \int_0^{x_c} |p(x)| dx + \int_0^{x_f} p(x) dx \right) \quad (37)$$

Using integration by parts on all three integrals gives

$$S[x] = \frac{i}{\hbar} \left([xp]_{x_i}^{x_f} - \int_{p_i}^0 x(p) dp - i(4n + 2) \int_0^\pi |x(p)|^2 d\varphi - \int_0^{x_f} x(p) dp \right) \quad (38)$$

Where the central integral became an integral along the semicircle in the complex p plane. Which means we can think about it as tunneling along the p axis, jumping from positive momentum to negative momentum in the same way that the trajectories in under the barrier tunneling jump from a to b .